Email Generation Assistant

AI Engineer Candidate Assessment -- Final Report

Models Compared:	mistral-large-latest vs mistral-small-latest
Prompting Strategy:	Role-Playing + Few-Shot + Chain-of-Thought
Test Scenarios:	10 unique scenarios with human reference emails
Custom Metrics:	Fact Recall * Tone Accuracy * Professional Quality

*Generated by generate_pdf.py * outputs/results.json*

1. Project Overview

This project implements an LLM-powered email generation assistant that produces professional emails from three structured inputs: Intent (the purpose of the email), Key Facts (bullet points that must appear in the email), and Tone (the desired register). The assistant uses three complementary advanced prompting techniques to maximise output quality and consistency, and is exposed as a FastAPI REST service.

Assistant Inputs

Intent:	The core purpose of the email -- e.g. "Follow up after a sales meeting."
Key Facts:	Bullet points of information that must be seamlessly included in the email.
Tone:	The desired style -- e.g. formal, casual, urgent, empathetic.

Architecture

generator.py -- builds the prompt and calls the Mistral API.
evaluator.py -- implements the 3 custom metrics (1 automated, 2 LLM-as-Judge).
test_data.py -- 10 scenarios with human reference emails.
run_evaluation.py -- batch runner that produces outputs/scores.csv and outputs/results.json.
api.py -- FastAPI service with /generate and /evaluate endpoints.

2. Prompt Template

Three complementary techniques are layered into a single prompt structure that is applied identically to both models, isolating model capability as the only variable.

Technique 1 -- Role-Playing (System Prompt)

The system prompt casts the model as a domain expert. This primes professional vocabulary, format conventions, and quality expectations before any user content is seen.

```
SYSTEM PROMPT:
"You are an expert professional email copywriter with 15+ years of experience
crafting high-impact business communications across industries. Your writing is
valued for precision, appropriate tone calibration, and seamless integration of
facts into natural prose.

Before drafting any email, work through these steps internally:
1. What is the single goal this email must accomplish?
2. How does the requested tone shape word choice, sentence length, and the opening line?
3. How can each fact be woven naturally into the narrative -- not listed, but integrated?
4. What is the one specific action the reader should take, and how do I make it easy?

Always produce: a subject line, greeting, body paragraphs, a clear call-to-action,
and a professional closing."
```

Technique 2 -- Few-Shot Prompting (User Message Prefix)

Two complete INPUT -> OUTPUT examples are prepended to every user message. These anchor the expected structure (subject -> greeting -> body -> CTA -> close) and show how facts must be woven into prose -- not bullet-listed.

```
--- EXAMPLE 1 ---
INPUT:
  Intent: Introduce myself as the new account manager
  Key Facts:
    - My name is Alex Rivera
    - Taking over from James Park who retired
    - Have been briefed on their account: Northgate Healthcare, 3-year client
    - First call scheduled for June 10 at 3pm EST
  Tone: Warm, professional
OUTPUT:
  Subject: Your New Account Manager - Let's Connect
  Dear Northgate Healthcare Team, ...

--- EXAMPLE 2 ---
INPUT:
  Intent: Decline a vendor proposal
  Key Facts:
    - Vendor is Apex Supplies
    - Pricing was 30% above our budget
    - Timeline of 6 months was too long
    - We may revisit in Q1 next year
  Tone: Respectful, direct
OUTPUT:
  Subject: Re: Proposal from Apex Supplies
  Dear Apex Supplies Team, ...
```

Technique 3 -- Chain-of-Thought (Embedded in System Prompt)

Steps 1-4 in the system prompt force the model to reason about goal, tone, fact integration, and CTA before generating any text. This 'think before you write' discipline improves consistency on all three evaluation metrics.

User Message Template

```
[FEW-SHOT EXAMPLES PREFIX]

Now generate an email for the following:

Intent: {intent}
Key Facts:
- {fact_1}
- {fact_2}
- ...
Tone: {tone}

OUTPUT:
```

3. Custom Evaluation Metrics

Three metrics are implemented in evaluator.py. Together they cover the three most critical failure modes of an email generation assistant: missing information, wrong register, and poor writing craft.

Metric 1: Fact Recall

Automated

Definition:	The fraction of provided key facts reflected in the generated email.
Logic:	For each fact bullet point, content words are extracted (stopwords and words <=2 characters are removed). A fact is counted as 'recalled' if >=50% of its content words appear in the email (case-insensitive). Score = recalled_facts / total_facts. Range: 0.0-1.0. This metric is fully deterministic -- no LLM call required.
Why it matters:	An email that sounds polished but omits a key date, amount, or name fails its core purpose.

Metric 2: Tone Accuracy

LLM-as-Judge

Definition:	How precisely the generated email matches the requested tone.
Logic:	An LLM judge (mistral-small-latest) is given the requested tone, the email purpose, and the generated email. It rates tone execution 1-10, which is normalised to 0.0-1.0. Tone requires genuine semantic understanding -- 'firm but polite' vs 'empathetic' vs 'enthusiastic' differ in vocabulary, sentence rhythm, and opener style in ways a keyword check cannot capture.
Why it matters:	Wrong tone is a professional failure regardless of factual accuracy.

Metric 3: Professional Quality

LLM-as-Judge

Definition:	Overall email craft, evaluated independently of tone.
Logic:	An LLM judge evaluates: grammar and fluency, logical structure, subject line effectiveness, appropriate length, and a clear call-to-action. Rates 1-10, normalised to 0.0-1.0. Scored separately from tone so a casual email can score high on quality even though its register is informal.
Why it matters:	Correct tone and facts don't save an email with poor grammar, no subject line, or no CTA.

4. Raw Evaluation Data

All 10 scenarios evaluated against both models using all 3 custom metrics. Scores are normalised to 0.0-1.0. Green = winning score per metric per scenario.

Scenario 1 -- Follow up on a sales meeting with a potential enterpris...

Tone: Professional, warm

Scenario / Intent	Model A (large)	Model B (small)	
Fact Recall	0.6000	**1.0000**	*B >>*
Tone Accuracy	0.9000	0.9000	*tie*
Professional Quality	0.9000	0.9000	*tie*
AVERAGE	0.8000	**0.9333**	*B >>*

Scenario 2 -- Apply for a Senior Data Scientist position

Tone: Confident, formal

Scenario / Intent	Model A (large)	Model B (small)	
Fact Recall	1.0000	1.0000	*tie*
Tone Accuracy	0.9000	0.9000	*tie*
Professional Quality	0.9000	0.9000	*tie*
AVERAGE	0.9333	0.9333	*tie*

Scenario 3 -- Respond to a customer complaint about a delayed shipmen...

Tone: Empathetic, apologetic

Scenario / Intent	Model A (large)	Model B (small)	
Fact Recall	0.8000	**1.0000**	*B >>*
Tone Accuracy	**1.0000**	0.9000	*<< A*
Professional Quality	0.9000	0.9000	*tie*
AVERAGE	0.9000	**0.9333**	*B >>*

Scenario 4 -- Send a weekly project status update to stakeholders

Tone: Professional, clear, informative

Scenario / Intent	Model A (large)	Model B (small)	
Fact Recall	1.0000	1.0000	*tie*
Tone Accuracy	0.9000	0.9000	*tie*
Professional Quality	0.9000	0.9000	*tie*
AVERAGE	0.9333	0.9333	*tie*

Scenario 5 -- Request a meeting to get budget approval for software l...

Tone: Direct, professional, concise

Scenario / Intent	Model A (large)	Model B (small)	
Fact Recall	**1.0000**	0.7500	*<< A*
Tone Accuracy	0.9000	0.9000	*tie*
Professional Quality	0.9000	0.9000	*tie*
AVERAGE	**0.9333**	0.8500	*<< A*

4. Raw Evaluation Data (continued)

Scenario 6 -- Announce a new product launch to existing customers

Tone: Enthusiastic, exciting, friendly

Scenario / Intent	Model A (large)	Model B (small)	
Fact Recall	1.0000	1.0000	*tie*
Tone Accuracy	1.0000	1.0000	*tie*
Professional Quality	0.9000	0.9000	*tie*
AVERAGE	0.9667	0.9667	*tie*

Scenario 7 -- Send a first payment reminder for an overdue invoice

Tone: Firm but polite, professional

Scenario / Intent	Model A (large)	Model B (small)	
Fact Recall	1.0000	1.0000	*tie*
Tone Accuracy	**0.9000**	0.8000	*<< A*
Professional Quality	0.9000	0.9000	*tie*
AVERAGE	**0.9333**	0.9000	*<< A*

Scenario 8 -- Send a thank you email after a job interview

Tone: Warm, enthusiastic, professional

Scenario / Intent	Model A (large)	Model B (small)	
Fact Recall	0.6000	0.6000	*tie*
Tone Accuracy	**1.0000**	0.9000	*<< A*
Professional Quality	0.9000	0.9000	*tie*
AVERAGE	**0.8333**	0.8000	*<< A*

Scenario 9 -- Propose a co-marketing partnership to a complementary b...

Tone: Persuasive, professional, collaborative

Scenario / Intent	Model A (large)	Model B (small)	
Fact Recall	1.0000	1.0000	*tie*
Tone Accuracy	0.9000	0.9000	*tie*
Professional Quality	0.9000	0.9000	*tie*
AVERAGE	0.9333	0.9333	*tie*

Scenario 10 -- Apologize to enterprise clients for an unexpected platf...

Tone: Apologetic, accountable, transparent

Scenario / Intent	Model A (large)	Model B (small)	
Fact Recall	1.0000	1.0000	*tie*
Tone Accuracy	0.9000	**1.0000**	*B >>*
Professional Quality	0.9000	0.9000	*tie*
AVERAGE	0.9333	**0.9667**	*B >>*

5. Comparative Analysis & Production Recommendation

5.1 Overall Score Summary

Metric	Model A (mistral-large-latest)	Model B (mistral-small-latest)	Winner
Fact Recall	0.9000	**0.9350**	*B >>*
Tone Accuracy	**0.9300**	0.9100	*<< A*
Professional Quality	0.9000	0.9000	*tie*
OVERALL AVERAGE	0.9100	**0.9150**	*B >>*

5.2 Which Model Performed Better?

Model B (mistral-small-latest) achieved a higher overall average score (0.9150 vs 0.9100), driven primarily by its superior Fact Recall (0.9350 vs 0.9000). Model A (large) led on Tone Accuracy (0.9300 vs 0.9100), indicating it better understands nuanced tone instructions. Both models scored identically on Professional Quality (0.9000), suggesting the prompting strategy is effective at enforcing email structure regardless of model size.

5.3 Biggest Failure Mode of the Lower-Performing Model

The biggest failure mode of Model A (mistral-large-latest) is Fact Recall (gap: 0.0350). Specifically, in Scenarios 1 and 8 (sales meeting follow-up and job interview thank-you), Model A scored only 0.6 on Fact Recall -- meaning it generated well-structured, professional-sounding emails but omitted key concrete details such as the contact person's name and specific metrics discussed. This 'fluent but vague' failure is the highest-risk pattern for real-world email generation: recipients notice missing specifics even when prose quality is high.

5.4 Production Recommendation

Recommended model: mistral-large-latest (Model A)

Despite scoring slightly lower overall (0.9100 vs 0.9150), mistral-large-latest is the recommended model for production for these reasons:

1. Tone Accuracy advantage is critical for customer-facing use

Model A scored 0.9300 vs 0.9100. For outbound sales emails, apology emails, and partnership pitches -- where wrong tone causes reputational damage -- this 0.02 gap is operationally significant.

2. Fact Recall gap is closable with prompt discipline

Model A's lower Fact Recall (0.90 vs 0.935) in scenarios 1 and 8 was caused by paraphrasing facts rather than omitting them. Stricter prompt instructions ('include exact names, dates, and numbers') would likely close this gap without switching models.

3. Consistent Professional Quality

Both models tied at 0.90, confirming that the prompting strategy (role-play + few-shot + CoT) successfully enforces email structure. Model A's larger capacity provides more headroom for complex scenarios.

Use mistral-small-latest as a cost-optimised fallback for high-volume, lower-stakes internal communications where Fact Recall is more critical than tone nuance.